FinOps INSIGHTS

CLOUD OPTIMIZATION REPORT

Optimize Resources. Reduce Waste. Maximize Value.

Comprehensive analysis of cloud costs, waste, and optimization opportunities across all projects and regions.

$517.54K	$179.36K	$184.62K	35.6%
Total Cost	Total Waste Cost	Est. Savings	Cost Optimization Potential

Discover
Identify underutilized and idle resources driving unnecessary costs.

Optimize
Rightsize, shutdown, and recommend optimal resource configurations.

Save
Reduce waste and improve efficiency to maximize cloud value.

Govern
Implement policies and monitor continuously for sustained optimization.

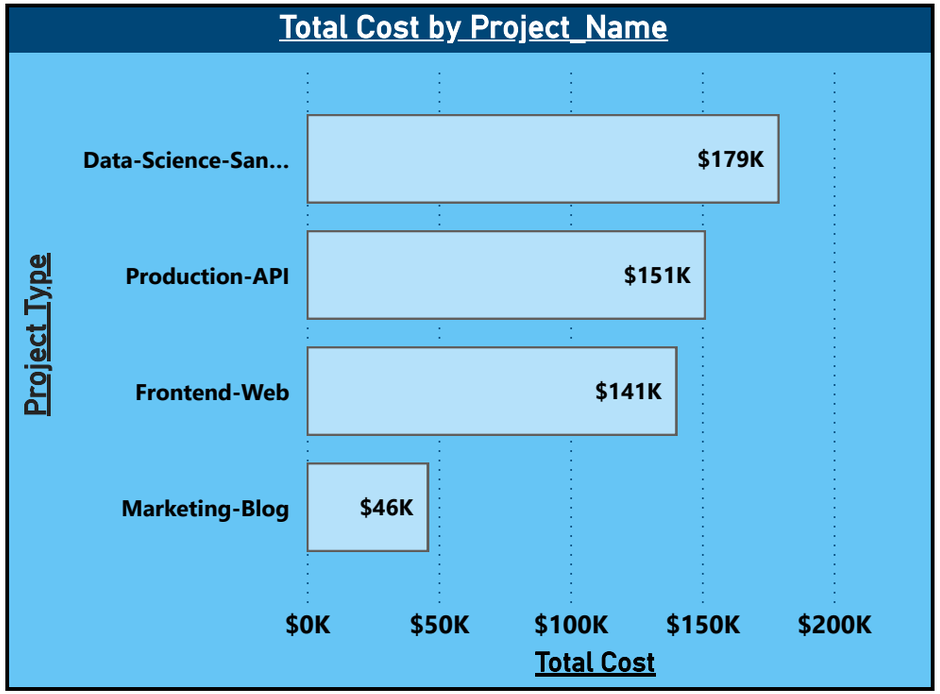
$517.54K	$179.36K	34.66%	100K	80K	20K	$500.84K
Total Cost	Total Waste Cost	Waste %	Total Instances	Active Instances	Idle Instances	Projected Monthly C...

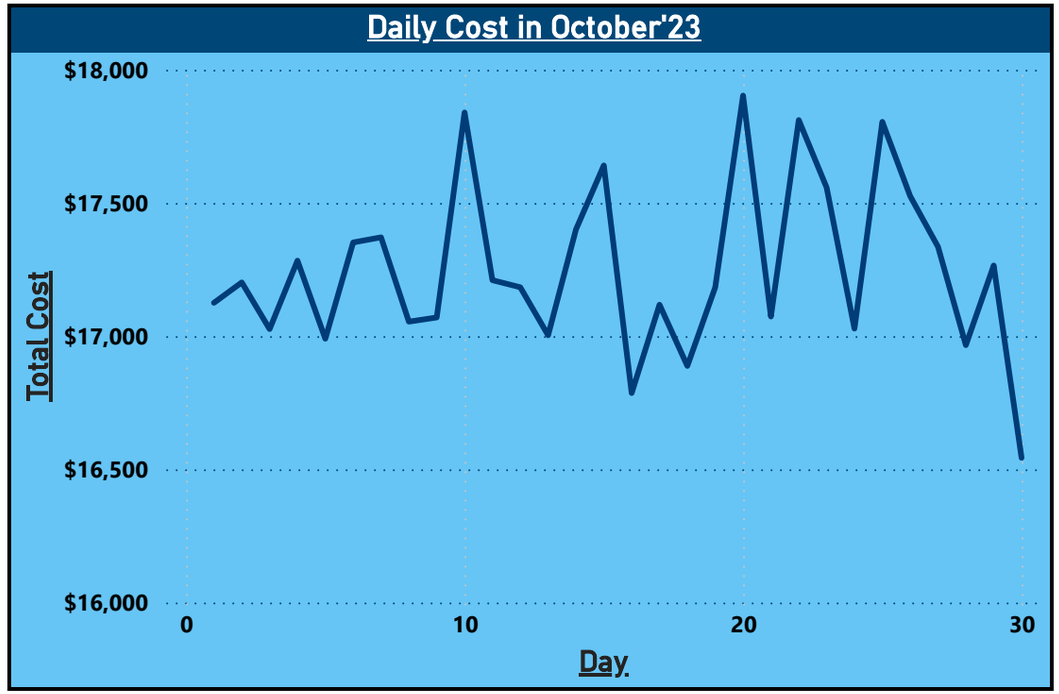
Region: All

- ap-south-1
- eu-west-1
- us-east-1

Instance_Type: All

- c5.xlarge
- m5.large
- r5.2xlarge
- t3.micro





- Out of **$517.54K** total cost, around **$179.36K (34.66%)** is **waste**, which is a significant optimization opportunity.
- **20K** out of **100K** instances are **idle (20%)**, contributing heavily to unnecessary costs.
- **Data-Science-Sandbox** and **Production-API** projects are the highest cost drivers, so optimization focus should be there first.
- **Data-Science-Sandbox** project has the **highest total cost (~$180K)**, indicating either heavy usage or poor optimization.
- Daily costs show **frequent spikes (~$17K–$18K)** with no stable pattern, suggesting inconsistent workload or an inefficient scaling.
- Here, Total Cost for this **October'23** is **$517.54K** but on an average of daily burn rate, its Projected Monthly Cost is about **$500.84K** for **30 days**.

$179.36K

Total Waste Cost

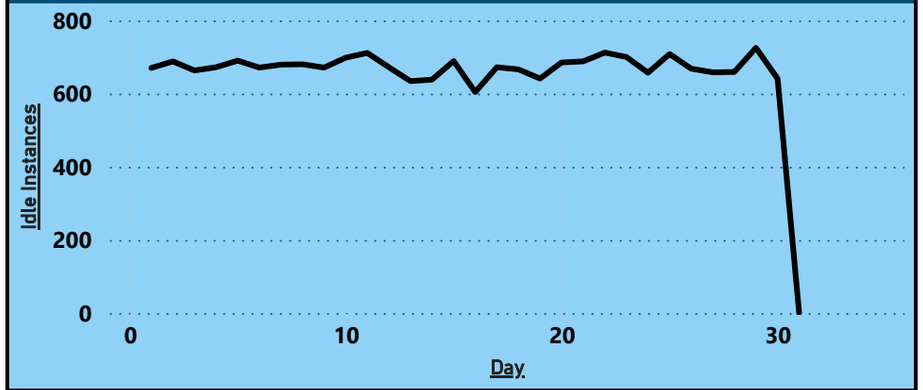
80K

Active Instances

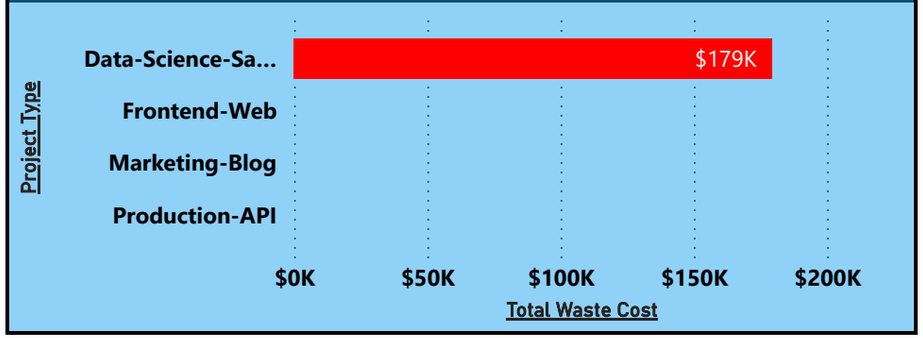
20K

Idle Instances

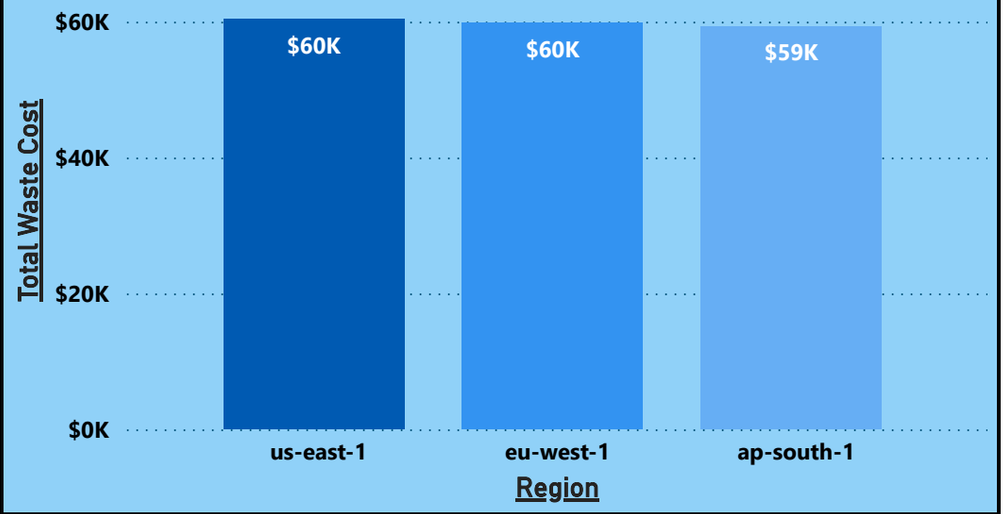
Idle Instances per Day in October'23



Total Waste Cost by Project Type



Total Waste Cost by Region



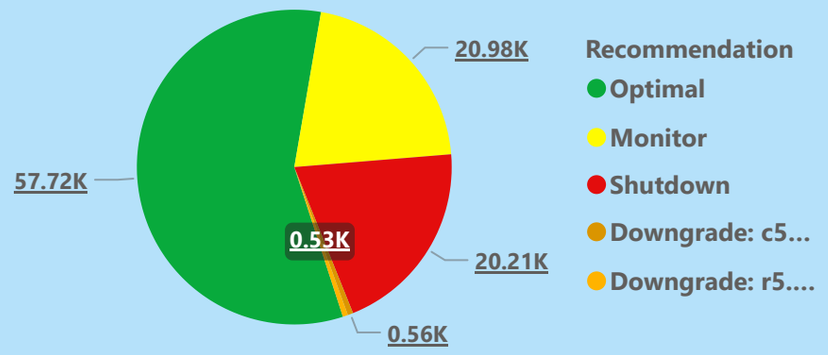
Idle Resource Flagger

Instance_ID	Project_Type	Region	CPU %	Daily_Cost	Uptime_Hours	Idle_Flag	Waste Cost
i-10011965	Data-Science-Sandbox	ap-south-1	5.04	$12.1	481	Idle	$12.10
i-10014881	Data-Science-Sandbox	us-east-1	2.60	$12.1	401	Idle	$12.10
i-10023976	Data-Science-Sandbox	ap-south-1	5.82	$12.1	230	Idle	$12.10
i-10027920	Data-Science-Sandbox	eu-west-1	5.60	$12.1	566	Idle	$12.10
i-10040650	Data-Science-Sandbox	us-east-1	1.44	$12.1	581	Idle	$12.10
i-10044597	Data-Science-Sandbox	us-east-1	1.26	$12.1	181	Idle	$12.10
i-10060859	Data-Science-Sandbox	ap-south-1	4.15	$12.1	446	Idle	$12.10
i-10085950	Data-Science-Sandbox	us-east-1	2.02	$12.1	126	Idle	$12.10
i-10093981	Data-Science-Sandbox	us-east-1	6.13	$12.1	160	Idle	$12.10

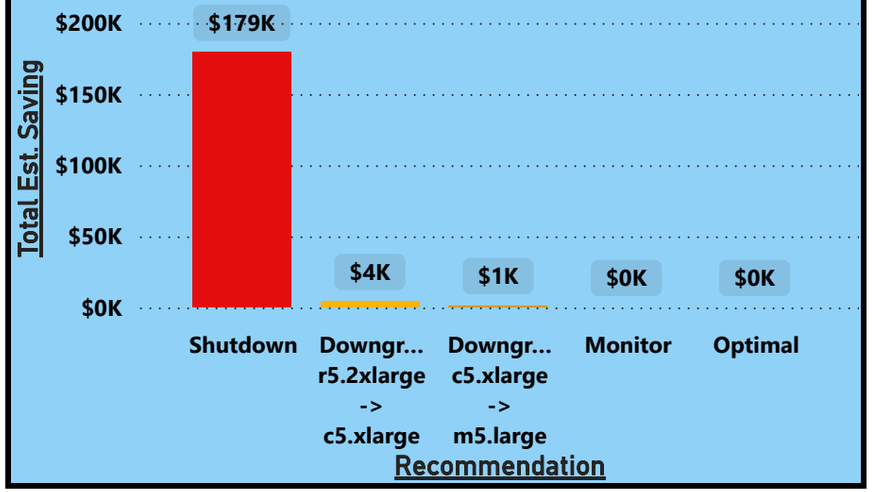
- The **Total Waste Cost (~$179K)** is almost completely attributed to **Data-Science-Sandbox**, confirming it as the core inefficiency source.
- Idle instances remain **~650–750 daily**, showing **persistent underutilization**, not a one-time spike → indicates poor resource scheduling or no auto-shutdown.
- Waste cost remains **consistently high across days**, meaning it's **predictable and recurring**, not random.
- Regions (**us-east-1, eu-west-1, ap-south-1**) all show **~$59K–$60K waste each**, meaning inefficiency is **globally distributed**, not region-specific.
- Many instances show **CPU < 10% with uptime > 72 hours**, yet still running → clear sign of **zombie instances consuming cost without value** which should be either **shutdown** or **downgrade.**

$517.54K	$179.36K	$184.62K	$332.92K
Total Cost	Total Waste Cost	Total Est. Savings	Optimized Cost

No. of Instances by Recommendation



Total Est. Saving by Recommendation



Recommendation Table

Instance ID	Project Type	Daily Cost	CPU %	Memory %	Idle Flag	Instance Type	Recommendation	Est. Saving
i-10000431	Data-Science-Sandbox	$4.08	7.00	13.53	Idle	c5.xlarge	Shutdown	$4.08
i-10010763	Data-Science-Sandbox	$4.08	2.77	8.31	Idle	c5.xlarge	Shutdown	$4.08
i-10011965	Data-Science-Sandbox	$12.1	5.04	8.30	Idle	r5.2xlarge	Shutdown	$12.1
i-10014881	Data-Science-Sandbox	$12.1	2.60	6.20	Idle	r5.2xlarge	Shutdown	$12.1
i-10016090	Data-Science-Sandbox	$4.08	3.92	6.65	Idle	c5.xlarge	Shutdown	$4.08
i-10016325	Data-Science-Sandbox	$4.08	5.14	7.41	Idle	c5.xlarge	Shutdown	$4.08
i-10023976	Data-Science-Sandbox	$12.1	5.82	11.10	Idle	r5.2xlarge	Shutdown	$12.1
i-10027920	Data-Science-Sandbox	$12.1	5.60	14.35	Idle	r5.2xlarge	Shutdown	$12.1
i-10029025	Data-Science-Sandbox	$4.08	6.88	12.51	Idle	c5.xlarge	Shutdown	$4.08

- Total Waste is about **$179.36K** because of **zombie instances** should be shut down and also around **$5.26K** would be save after **downgrading** the instance so Est. Saving would be around **$184.62K.**
- Because of it, Cost reduces from **$517K → ~$333K**, showing **strong FinOps impact.**
- Also, Around **57–58% instances are "Optimal"**, meaning good resource allocation for more than half of the instances.
- Here we can see that **Shutdown** alone contributes the **largest portion (~$170K+) of savings**, far higher than downgrade.
- Here Downgrade from **r5.2xlarge -> c5.xlarge** is about **532 instances** and from **c5.xlarge -> m5.large** is about **557 instances**.

Cloud Cost Optimization & FinOps Analysis Summary Report

1. Raw Data Overview

- **Dataset:** Cloud FinOps Logs
- **Fields included:** Instance ID, Project Name, CPU Utilization %, Memory Utilization %, Uptime Hours, Daily Cost, Hourly Rate, Region, Instance Type
- **Data Range:** 1 Oct – 30 Oct (Daily Level Data)
- **Problem:** Raw data had no clear classification of idle/waste/optimization

2. Data Cleaning & Transformation

- Removed duplicates and null values
- **Standardized:**
  - Region names (ap-south, Eu-west, us-east)
  - Instance types (micro → 2xlarge)
- **Converted:** Date formats for time analysis
- Created structured dataset for **Power BI**

3. Data Preparation

Calculated Columns

- **Idle Flag:** IF CPU < 10% AND Uptime > 72 hours → "Idle", ELSE "Active"
- **Daily Waste Cost:** IF Idle → Daily Cost ELSE 0
  - **Recommendation** Idle → Shutdown
  - Low CPU Utilization → Downgrade
  - Else → Keep

DAX Measures

- **Total Cost** = SUM(Daily Cost)
- **Total Waste Cost** = SUM(Daily Waste Cost)
- **Waste %** = Waste / Total Cost
- **Total Instances** = COUNT(Instance ID)
- **Idle Flag** = Active Instances / Idle Instances
- **Estimated Savings** = Based on recommendation logic
- **Optimized Cost** = Total Cost – Est. Savings

4. Dashboard Pages Overview

Page 1: Cost Overview

- Overall cost, waste, utilization
- Project-wise cost distribution
- Daily cost trend

Purpose: Understand where money is going

Page 2: Waste Analysis

- Idle instances trend
- Waste by region & project
- Detailed idle instance table

Purpose: Identify inefficiencies (root cause)

Page 3: Optimization & Savings

- Recommendation breakdown (Keep / Shutdown / Downgrade)
- Estimated savings
- Optimized cost projection

Purpose: Show action plan & impact

5. KPIs Used

- Total Cost
- Total Waste Cost
- Waste %
- Total Instances
- Active vs Idle Instances
- Estimated Savings
- Optimized Cost

6. Charts & Visuals Used

- **Bar Chart** → Cost by Project
- **Line Chart** → Daily Cost / Idle Trend
- **Pie Chart** → Recommendation Distribution
- **Column Chart** → Waste/Savings by Category
- **Table** → Instance-level analysis
- **Cards (KPI)** → Key metrics at top

Conclusion:

*By analyzing usage patterns and identifying idle & over-provisioned resources, we transformed raw cloud data into actionable insights, enabling **~35%** cost optimization through targeted shutdown and rightsizing strategies.*